

Mathematics Bootcamp

Functions, Counting, The Real Numbers

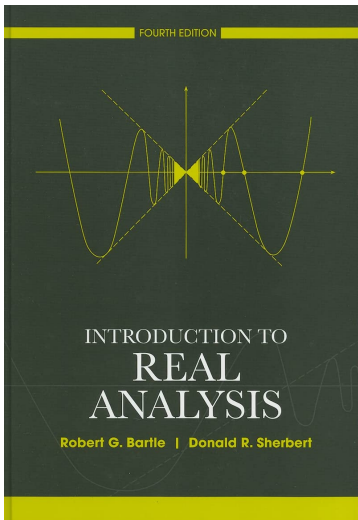
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September 2025

Today

- Cartesian Products and Functions
- Finite and Infinite Sets
- The Real Numbers and Euclidean Distance
- Sequences, Continuity, and Compactness
- Bounds, Completeness, and Ordinal Utility

Bartle-Sherbert - Real Analysis



- Most of what follows is taken from Bartle-Sherbert.
- My favorite introductory book for Graduate Level Math.
 - Careful attention to detail.
 - Great pedagogical order.

Cartesian Product

1.1.5 Definition

If A and B are nonempty sets, then the **Cartesian product** $A \times B$ of A and B is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. That is,

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

For example, if $A := \{x \in \mathbb{R} : 1 \leq x \leq 2\}$ and $B := \{y \in \mathbb{R} : 0 \leq y \leq 1 \text{ or } 2 \leq y \leq 3\}$, then instead of a rectangle, we should have a drawing such as Figure 1.1.3.

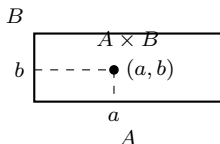


Figure 1.1.2

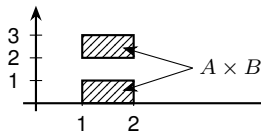


Figure 1.1.3

Functions

1.1.6 Definition

Let A and B be sets. Then a **function** from A to B is a set f of ordered pairs in $A \times B$ such that for each $a \in A$ there exists a unique $b \in B$ with $(a, b) \in f$. (In other words, if $(a, b) \in f$ and $(a, b') \in f$, then $b = b'$.)

The set A of first elements of a function f is called the **domain** of f and is often denoted by $D(f)$. The set of all second elements in f is called the **range** of f and is often denoted by $R(f)$. Note that, although $D(f) = A$, we only have $R(f) \subseteq B$. (See Figure 1.1.4.)

When $A, B \subseteq \mathbb{R}$, uniqueness is the *vertical line test*: the graph meets each line $x = a$ at most once. Since $D(f) = A$, it meets that line exactly once for every $a \in A$.

Functions as Graphs

- A function f maps a to *exactly one* b : $f(a) = b$ or $a \mapsto b$

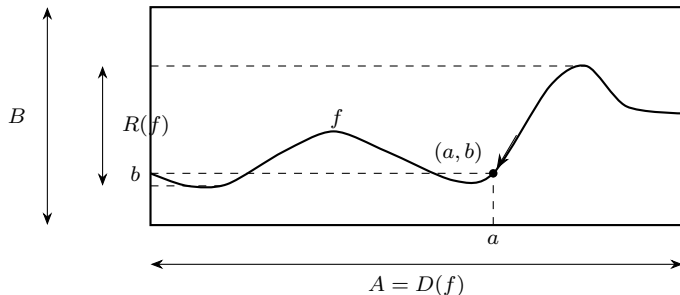


Figure 1.1.4 A function as a graph

Composition and Inverse Images

Composition

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. Their composition is the function

$$g \circ f : X \rightarrow Z, \quad (g \circ f)(x) := g(f(x)).$$

The order matters: apply f first and then apply g .

Inverse Image

For a set $D \subseteq Y$, the inverse image of D under f is

$$f^{-1}(D) := \{x \in X : f(x) \in D\}.$$

The notation $f^{-1}(D)$ is defined for every function; f need not be bijective.

Injective, Surjective, and Bijective

1.1.9 Definition

Let $f : A \rightarrow B$ be a function from A to B .

- (a) The function f is said to be **injective** (or **one-to-one**) if whenever $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$. If f is an injective function, we also say that f is an **injection**.
- (b) The function f is said to be **surjective** (or to map A **onto** B) if $f(A) = B$; that is, if the range $R(f) = B$. If f is a surjective function, we also say that f is a **surjection**.
- (c) If f is both injective and surjective, then f is said to be **bijective**. If f is bijective, we also say that f is a **bijection**.

Proofs - Injective and Surjective

- In order to prove that a function f is injective, we must establish that:

for all x_1, x_2 in A , if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

To do this we assume that $f(x_1) = f(x_2)$ and show that $x_1 = x_2$.

- To prove that a function f is surjective, we must show that for any $b \in B$ there exists at least one $x \in A$ such that $f(x) = b$.

Cardinality - Finite and Infinite Sets

1.3.1 Definition

- (a) The empty set $\emptyset$ is said to have 0 **elements**.
- (b) If $n \in \mathbb{N}$, a set S is said to have n **elements** if there exists a bijection from the set $\mathbb{N}_n := \{1, 2, \dots, n\}$ onto S .
- (c) A set S is said to be **finite** if it is either empty or it has n elements for some $n \in \mathbb{N}$.
- (d) A set S is said to be **infinite** if it is not finite.

1.3.6 Definition

- (a) A set S is said to be **denumerable** (or **countably infinite**) if there exists a bijection of $\mathbb{N}$ onto S .
- (b) A set S is said to be **countable** if it is either finite or denumerable.
- (c) A set S is said to be **uncountable** if it is not countable.

Infinite Sets Can Have Different Cardinalities

Comparing Sizes

Two sets have the same cardinality when there is a bijection between them. This definition applies to both finite and infinite sets.

- $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ are countably infinite: each can be put in bijection with $\mathbb{N}$.
- $\mathbb{R}$ is uncountable: there is no bijection $\mathbb{N} \rightarrow \mathbb{R}$.

Cantor's diagonal argument proves the second claim: video.

Why the Real Numbers Matter

- Economic alternatives often lie in $\mathbb{R}^L$, while utility is real-valued.
- We need the structure of $\mathbb{R}$, not a construction of it.
- Three ingredients will matter:
 - order, for comparisons;
 - completeness, for limits and extrema;
 - distance, for neighborhoods and convergence.

$\mathbb{R}$ Is a Complete Ordered Field

Ordered Field

A **field** supports addition and multiplication satisfying the field axioms. A **total order** compares every pair of elements and is compatible with the field operations.

$$\begin{aligned}a < b &\implies a + c < b + c, \\ 0 < a \text{ and } 0 < b &\implies 0 < ab.\end{aligned}$$

Totality gives **trichotomy**: for any a, b , exactly one of $a < b$, $a = b$, or $a > b$ holds.

The Structural Difference

Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. Of these two, only $\mathbb{R}$ is **complete**: every nonempty set of real numbers bounded above has a least upper bound in $\mathbb{R}$.

Absolute Value and the Real Line

2.2.1 Definition

For $a \in \mathbb{R}$, its **absolute value**, denoted $|a|$, is

$$|a| := \begin{cases} a, & \text{if } a > 0, \\ 0, & \text{if } a = 0, \\ -a, & \text{if } a < 0. \end{cases}$$

The Real Line

A convenient and familiar geometric interpretation of the real number system is the real line. In this interpretation, the absolute value $|a|$ of an element a in $\mathbb{R}$ is regarded as the distance from a to the origin 0. More generally, the **distance** between elements a and b in $\mathbb{R}$ is $|a - b|$. (See Figure 2.2.3.)

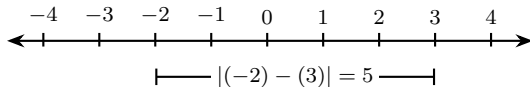


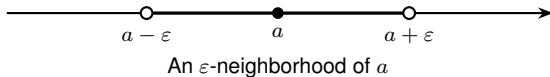
Figure 2.2.3 The distance between $a = -2$ and $b = 3$

Neighborhoods in $\mathbb{R}$ and $\mathbb{R}^L$

2.2.7 Definition

For $a \in \mathbb{R}$ and $\varepsilon > 0$, the ε -**neighborhood** of a is

$$V_\varepsilon(a) := \{x \in \mathbb{R} : |x - a| < \varepsilon\} = (a - \varepsilon, a + \varepsilon).$$



Euclidean Neighborhood

For $x \in \mathbb{R}^L$ and $\varepsilon > 0$,

$$B_\varepsilon(x) := \{y \in \mathbb{R}^L : \|y - x\| < \varepsilon\}, \quad \|z\| := \sqrt{\sum_{\ell=1}^L z_\ell^2}.$$

The Triangle Inequality Controls Distances

Triangle Inequality

For every $x, y, z \in \mathbb{R}^L$,

$$\|x - z\| \leq \|x - y\| + \|y - z\|.$$

Geometric Meaning

The direct distance from x to z cannot exceed the distance obtained by travelling from x to z through an intermediate point y .

Sequence Convergence in $\mathbb{R}^L$

Convergence

A sequence (x_n) in $\mathbb{R}^L$ **converges** to $x \in \mathbb{R}^L$ if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N, \quad \|x_n - x\| < \varepsilon.$$

We write $x_n \rightarrow x$. Equivalently, every neighborhood of x contains all terms of the sequence from some point onward.

Coordinatewise Convergence

For $x_n = (x_{n1}, \dots, x_{nL})$ and $x = (x_1, \dots, x_L)$,

$$x_n \rightarrow x \quad \Longleftrightarrow \quad x_{n\ell} \rightarrow x_\ell \quad \text{for every } \ell = 1, \dots, L.$$

Continuity of a Function

Sequential Definition

Let $X \subseteq \mathbb{R}^L$, $Y \subseteq \mathbb{R}^K$, and $f : X \rightarrow Y$. The function f is *continuous at* $x \in X$ if, for every sequence (x_n) in X ,

$$x_n \rightarrow x \implies f(x_n) \rightarrow f(x).$$

It is continuous on X if it is continuous at every $x \in X$.

Keep the Objects Separate

This is continuity of the function f . Day 3 defines continuity of a preference relation through closed upper and lower contour sets.

Closed and Bounded Sets Are Compact in $\mathbb{R}^L$

Heine–Borel and Extreme Values

Let $D \subseteq \mathbb{R}^L$. Then

$$D \text{ is compact} \iff D \text{ is closed and bounded.}$$

If D is nonempty and compact and $f : D \rightarrow \mathbb{R}$ is continuous, then there exist $x_{\min}, x_{\max} \in D$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \text{for every } x \in D.$$

Compactness controls the feasible set; continuity controls the objective. Together they turn a supremum or infimum into an attained maximum or minimum.

Bounds

2.3.1 Bounds

Let $\emptyset \neq S \subseteq \mathbb{R}$.

u is an upper bound $\iff s \leq u$ for every $s \in S$,

w is a lower bound $\iff w \leq s$ for every $s \in S$.

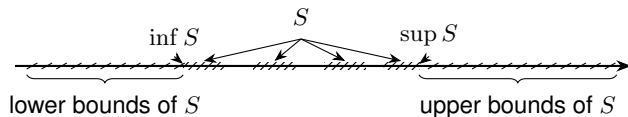
S is **bounded** if it has both an upper and a lower bound.

Example

For $S = (0, 1)$,

upper bounds $= [1, \infty)$, lower bounds $= (-\infty, 0]$.

Suprema and Infima



2.3.2 Supremum = least upper bound

$$u = \sup S$$

1. $s \leq u$ for every $s \in S$.
2. $u \leq v$ for every upper bound v .

$u \in S \Rightarrow u = \max S$;
 $u \notin S \Rightarrow$ no maximum.

Infimum = greatest lower bound

$$w = \inf S$$

1. $w \leq s$ for every $s \in S$.
2. $t \leq w$ for every lower bound t .

$w \in S \Rightarrow w = \min S$;
 $w \notin S \Rightarrow$ no minimum.

Completeness of the Real Numbers

2.3.6 Completeness Property of $\mathbb{R}$

For every nonempty $S \subseteq \mathbb{R}$,

$$S \text{ bounded above} \implies \sup S \in \mathbb{R}.$$

Consequently, if S is bounded below, then $\inf S \in \mathbb{R}$.

Why $\mathbb{Q}$ Is Not Complete

$$A = \{q \in \mathbb{Q} : q^2 < 2\}$$

is bounded above, but its supremum in $\mathbb{R}$ is $\sqrt{2} \notin \mathbb{Q}$. Thus A has no supremum in $\mathbb{Q}$.

Completeness means that $\mathbb{R}$ has no gaps.

Key Implications for Proofs

Archimedean Property

2.4.3 *If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.*

2.4.5 Corollary *For every $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $1/n < \varepsilon$.*

Density Results

2.4.8 Density Theorem *If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.*

2.4.9 Corollary *If x and y are real numbers with $x < y$, then there exists an irrational number z such that $x < z < y$.*

Utility Is Ordinal

Ordinal Utility

A function $u : X \rightarrow \mathbb{R}$ represents $\succeq$ if

$$x \succeq y \iff u(x) \geq u(y).$$

If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then $\phi \circ u$ represents the same preferences. Utility levels and differences have no intrinsic meaning.

Existence

Every utility representation induces rational preferences. On finite or countable X , rationality is also sufficient. On the standard Euclidean choice set $X = \mathbb{R}_+^L$, a rational and continuous preference relation admits a continuous representation; Day 3 states the theorem.

Back to Economics - Utility Functions

Utility Functions

In economics, we often describe preference relations by means of a *utility function*. A utility function $u(x)$ assigns a numerical value to each element in X , ranking the elements of X in accordance with the individual's preferences. This is stated more precisely in Definition 1.B.2.

Definition 1.B.2: A function $u: X \rightarrow \mathbb{R}$ is a *utility function representing preference relation* $\succsim$ if, for all $x, y \in X$,

$$x \succsim y \Leftrightarrow u(x) \geq u(y).$$

Utility Representation Implies Rationality

The ability to represent preferences by a utility function is closely linked to the assumption of rationality. In particular, we have the result shown in Proposition 1.B.2.

Proposition 1.B.2: A preference relation $\succsim$ can be represented by a utility function only if it is rational.

Utility Generates the Same Choice from Every Menu

Representation Preserves Maximizers

Suppose $u : X \rightarrow \mathbb{R}$ represents $\succeq$. For every nonempty menu $B \subseteq X$ on which a best alternative exists,

$$C_{\succeq}(B) = \{x \in B : x \succeq y \text{ for every } y \in B\} = \arg \max_{x \in B} u(x).$$

If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then

$$\arg \max_{x \in B} u(x) = \arg \max_{x \in B} (\phi \circ u)(x).$$

Only the ranking matters. A strictly increasing transformation can simplify the objective without changing the chosen alternatives.